Category 1	Category 2	The middle Category	Category 4	Programming Questions
100 The value "2004.98" is a: a) Floating-point number (float) b) Integer c) String d) Decimal c) String	**100** What is an informal English-like language used for "thinking out" or outlining an algorithm? Pseudocode	**100** True or False: You can nest an if statement within a for loop, but you cannot nest a for loop within an if statement. False.	**100** True or False: The statement val = val ** 2 may be replaced by the more concise augmented assignment statement val *= 2. False	**100** You will have two minutes for this question. Open up Cloud9 and write a Python which asks the user how old they are. Then output how old they will be on their next birthday. Your program would output the following: Enter your age: 39 On your next birthday you will be 40. # Ask the user to enter their age as input age = int(input("Enter your age: ")) # Calculate the age on the next birthday next_birthday_age = age + 1

				# Output the age on the next birthday print("On your next birthday you will be", next_birthday_age, ".")
200 What is returned if we run the following code: print("total:", 9 ** (1/2)) ..... total: 3.0	**200** What is the value of the following expression: "20" + "45" ..... "20" + "45" = "2045"	**200** What does the following for statement print? for counter in range(17): print(counter, end=' ') 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16	**200** True or False: Something in a while statement must ensure that the condition eventually becomes False. Otherwise, a logic error called an infinite loop occurs. ..... ~~True~~ This answer is not correct. Other factors may end a while loop such as a break statement	**200** You will have two minutes to answer this question. Open up Cloud9 and write a Python code that will take a user's three favorite colors. Save each of these colors to a list and print them to the console with a for loop. ..... # Initialize an empty list to store favorite colors favorite_colors = [] # Use a for loop to prompt the user for their three favorite colors for i in range(3): color = input("Enter your favorite color: ")

				favorite_colors. append(color) # Add the color to the list # Use a for loop to print the favorite colors print("Your favorite colors are:") for color in favorite_colors: print(color)
300 List three reasons why Python is such a popular programming language: ..... 1. It is open source 2. It is easy to read 3. It is widely available to anyone	**300** What is the eval() function attempting to do in the following code: age = eval(input("What is your age? ")) print("You are", age, "years old.") print("Next year, you will be", age + 1, "years old.") ..... The eval() function attempts to take the user input and change it into an integer so a mathematic operator can be	**300** True or False: In Python, it is possible to have multiple elif blocks after an if statement without having an else block. ..... True	**300** True or False: Lists can technically store heterogeneous data, but it is better practice and more useful to use them for only homogeneous data. ..... True.	**300** You will have 3 minutes for this question. Open up Cloud9 and write a Python program that outputs the following numbers, with a space separating each number, in a for loop: Note: 18 is skipped (not outputted). You can use whatever logic you like, to skip 18. 2 4 6 8 10 12 14 16 20

	used on it in the print statement.			# Use a for loop to iterate through the range of numbers from 2 to 20 (inclusive)
				
				for i in range(2, 21):
				
				# Check if the current number is not equal to 18
				
				if i != 18:
				
				# Print the current number followed by a space
				
				print(i, end=" ")
				
				# Output: 2 4 6 8 10 12 14 16 20

400 What does it mean that Python is a case-sensitive language? It means that lowercase and uppercase characters matter, especially when naming variables. ie. Number != number	**400** What does the following input/output demonstrate: In [1]: x = 7 Out[1]: int In [2]: type(x) Out[2]: int In [3]: x = 4.1 In [4]: type(x) Out[4]: float In [5]: x = 'dog'	**400** In regards to the following code: def add_two_nums(num1, num2): return num1 + num2 If a non-integer number is entered into the function call (ie, add_two_nums(1, "apple pie"), what will likely happen?	**400** You have a variable total initialized to 1. You want to repeatedly double the value of total until it becomes greater than 100. Which of the following while loop conditions achieves this goal? A) while total >= 100:	**400** You will have 3 minutes for this question. Open up Cloud 9 and write a program that outputs the following numbers, with comma and space separating each number, in a for loop:

	In [6]: type(x) Out[6]: str Over its lifetime a variable can be changed to different data types. Python uses dynamic typing–it determines the type of the object a variable refers to while executing your code. Python creates objects in memory and removes them from memory as necessary. This removal process–called garbage collection–helps ensure that memory is available for new objects you create.	We will get a TypeError because "apple pie" is a string and cannot be added with an integer when the operator + is being used in that way	B) while total < 100: C) while total > 100: D) while total <= 100: B) while total < 100	1, 10, 2, 20, 3, 30, 4, 40, 5, 50 # Use a for loop to iterate through the range of numbers for i in range(1, 6): # Print the current number and its multiple of 10 print(i, i * 10, end=", ") # end=", " adds a comma and space between numbers # Output: 1, 10, 2, 20, 3, 30, 4, 40, 5, 50,

500
Explain the difference between floor division and float division. Provide the operator for both, and estimate what the following expression will total:
 	
-20 // 6
	500
	True or False:
	
	The following code creates a slice consisting of the elements at indices 4 through 7 of the list numbers:
	
	numbers = [2, 5, 9, 8, 34, 42, 54, 89]
	numbers2 = numbers[4:7]
		500
		Do code snippet 1 and code snippet 2 output the same or different answers?
		
		#code snippet 1
		total = 200
		total /= 20
		print(total)
		
		#code snippet 2
		total = 200
		total //= 20
		print(total)
			500
			Consider the following code:
			
			def mystery_functio n(n):
			
			if n == 1:
			
			return 1
			
			else:
			
			return n * mystery_functio n(n - 1)
				500
				You have three minutes for this question.
				
				Open up Cloud9 and write the Python code which will calculate a 3 day moving average for the list below, in a for loop.
				
				I want to see good logic, and I am not so

/ = float division. Will always return a float value // = floor division. Will always round down to the nearest integer (can sometimes return as a float but always rounded) -20 // 6 = -4	False	They will output the same ish answer which is 10. (I say ish because one will return 10 and the other will return 10.0)	result mystery_functio n(5) What will the output be? A) 5 B) 10 C) 15 D) 20 B) 120	worried about syntax and if your code runs or not. lst = [1,2,3,4,5,6,7,8,9] lst = [1, 2, 3, 4, 5, 6, 7, 8, 9] moving_averages = [] # Step 1: Initialize an empty list for moving averages # Step 2: Iterate through the original list using a for loop for i in range(len(lst) - 2): # Step 3: Calculate the 3-day moving average for each index i three_day_avg = (lst[i] + lst[i + 1] + lst[i + 2]) / 3 # Step 4: Append the calculated average to the list of moving averages moving_averages.append(three_day_avg)

				print("3-day moving averages:", moving_averages)

Additional Topics

- Git
  - git add
  - git commit -m " "
  - git push
- Pandas
  - .loc/.iloc
  - indexing a dataframe